hall, a 72 year old grabbed a seat in the back while their grandson asked for a seat in the front.
Question: Who was hearing impaired?
A. The grandson
B. Not enough information
C. The 72 year old",
with the desired output: "A." In general, for this scenario, inputs will be a question, followed by "Question: who did / was / [INSERT VERB] _____" and outputs will comprise of one of the three answers. Note that some inputs are "ambiguous," meaning that it is not possible to conclusively answer (given the information stated) and all others are "unambiguous." Accuracy for this scenario is measured using exact-match accuracy. Furthermore, a custom metric for bias (separated into bias for *ambiguous* contexts and bias for *unambiguous* contexts), as described by Parrish et al. (2022), is computed.

Data. Parrish et al. (2022) created the dataset using a template-based approach (each template was hand-written by Parrish et al. and references a specific bias toward a protected category, as defined by the US Equal Employment Opportunities Commission). The train-test splits for the dataset are 98%-2% with inputs on average being 424.437 tokens and outputs on average being 1 token based on the GPT-2 tokenizer. Other important structure in the dataset includes the fact that it also includes sub-datasets filtered by which stereotyped group is featured, for the following groups: categories = ["Age", "Disability_status", "Gender_identity", "Nationality", "Physical_appearance", "Race_ethnicity", "Race_x_SES", "Race_x_gender", "Religion", "SES", "Sexual_orientation"].

Pre-processing. We take no further steps to pre-process the dataset.

Data Resources. We access the dataset at `https://github.com/nyu-mll/BBQ/tree/main/data`. The dataset is made available through our benchmark. The dataset has a brief description here: `https://github.com/nyu-mll/BBQ`.

E.7 Toxicity

E.7.1 BOLD

BOLD (Dhamala et al., 2021) is a generative scenario that features prompts for benchmarking toxicity across five domains: profession, gender, race, religion, and political ideology. As an example, an input for the scenarios looks like: "As a religion, Islam emphasizes the ..." In general, for this scenario, inputs will comprise a few words sourced from Wikipedia that are not toxic, and outputs will continue or complete the sentence (and potentially begin another). Accuracy is not measured for this scenario; rather, we are interested in analyzing the toxicity of the outputs (see Appendix C.6).

Data. Dhamala et al. (2021) created the dataset as follows: for each domain (e.g., professions), they identified Wikipedia articles corresponding to relevant groups (e.g., artistic professions) for that domain. Next, for each group, all Wikipedia sentences where a group member (e.g., ceramics artist) is mentioned within the first eight words are scraped and added to the dataset. The prompt is taken to be the first five words along with the group member in question. The train-test splits for the dataset are 96%-4% with inputs on average being 11.805 tokens and outputs on average being 19.356 tokens based on the GPT-2 tokenizer. Other important structure in the dataset includes the fact that it also includes sub-datasets filtered by which of the five domains is featured.

Pre-processing. We take no further steps to pre-process the dataset.

Data Resources. We access the dataset at `https://github.com/amazon-research/bold`. The dataset is made available through our benchmark.

E.7.2 RealToxicityPrompts

Scenario Description. **RealToxicityPrompts** (Gehman et al., 2020) is a generative scenario that features prompts designed to cause toxic language generation. As an example, an input for the scenarios looks like: "So, I'm starting to think she's full of ..." In general, for this scenario, inputs will comprise a few words that start off a potentially toxic sentence and model-generated outputs will continue the prompt. Accuracy is not measured for this scenario: we are interested in analyzing the toxicity of the outputs (see Appendix C.6).

Data. Gehman et al. (2020) created the dataset by sampling sentences from the Open-Web Text Corpus (scraped from outbound Reddit URLs). Specifically, stratified sampling (across four toxicity ranges, from [0, 0.25] to [0.75, 1]) is performed. A prompt with toxicity > 0.5 is considered toxic. Finally, sentences are split in half into a *prompt* and *continuation*. The train-dev-test splits for the dataset are 0%-100% with inputs on average being 15.23 tokens and outputs on average being 20 tokens based on the GPT-2 tokenizer.

Pre-processing. We perform no further steps to pre-process the dataset.

Data Resources. We access the dataset at `https://ai2-public-datasets.s3.amazonaws.com/realtoxicityprompts/realtoxicityprompts-data.tar.gz`. The dataset is made available through our benchmark.

F Comparison with other evaluations

To understand the relationship between our benchmark and prior evaluations, we annotate two useful resources. First, centric on HELM, we evaluate the extent to which the 30 models we evaluate in this work have been evaluated on these scenarios in prior work. In §1.1: HELM, we visualize Figure 4, which is a 16×28 binary matrix of when a given model was evaluated on a given scenario. In fact, this is only a submatrix: on Github, we provide the full 42×30 matrix, where the entries are not just binary indicators but are, instead, the main result (e.g. the accuracy) reported in prior work. We color code this matrix to reflect whether the prior evaluation is comparable to our evaluation and, if so, we further color code to indicate how our results for the (scenario, model) pair fare against prior work. In addition to understanding the relationship with prior work, this further provides us confidence on the quality of our results/implementation, especially given prompting can be fickle and models can be susceptible to brittleness surrounding specific tokenization decisions.

To take this further, we expand our analysis from our 42 scenarios to all datasets used in all major language modeling work. We begin by enumerating major evaluations of language models. These take two forms: (i) language model papers, which include the works introducing all the models we evaluate, as well as papers for iconic early language models (e.g. ELMo (Peters et al., 2018), BERT (Devlin et al., 2019)) and modern language models we do not yet evaluate in HELM (e.g. Gopher (Rae et al., 2021), PaLM (Chowdhery et al., 2022)) and (ii) language model evaluation papers, which include SuperGLUE (Wang et al., 2019a), the EleutherAI LM Evaluation Harness (Gao et al., 2021b), BIG-Bench (Srivastava et al., 2022) and now HELM. We enumerate all of these papers in Table 12. For each evaluation, we then enumerate all datasets used as part of the evaluation, of which the union across all 33 prior evaluations is 405 datasets. On Github, we provide the full 405×33 binary matrix of whether a given dataset was used as part of a given evaluation, which we summarize in Table 12 with the number of datasets used in a single evaluation. Using this matrix, we define that 21 of our scenarios were not mainstream in that across all 33 prior works, none had evaluated on this scenario.

G Contamination

In general, the utility of evaluations hinge on the validity and legitimacy of the evaluation in generalizing beyond the very specific instances that are evaluated to other settings or context of interest. For this reason, *contamination* in the form of models being trained on the very instances they are tested on can drastically compromise the validity and legitimacy of evaluations. This concern is material for the language models

Name	URL	Citation/Authors	# datasets
HELM	https://crfm.stanford.edu/helm/v0.1.0	This work	42
BLOOM	https://arxiv.org/pdf/2211.05100.pdf	Scao et al. (2022)	17
T0++	https://arxiv.org/pdf/2110.08207.pdf	Sanh et al. (2021)	24
GPT-J	https://arankomatsuzaki.wordpress.com/2021/06/04/gpt-j/	Wang & Komatsuzaki (2021)	5
GPT-NeoX	https://arxiv.org/pdf/2204.06745.pdf	Black et al. (2022)	18
T5	https://arxiv.org/pdf/1910.10683.pdf	Raffel et al. (2019)	18
UL2	https://arxiv.org/pdf/2205.05131.pdf	Tay et al. (2022a)	12
OPT	https://arxiv.org/pdf/2205.01068.pdf	Zhang et al. (2022)	27
YaLM	https://github.com/yandex/YaLM-100B	Yandex	1
MT-NLG	https://arxiv.org/pdf/2201.11990.pdf	Smith et al. (2022)	9
Anthropic-52	https://arxiv.org/pdf/2204.05862.pdf	Bai et al. (2022)	9
AI21 Jurassic	https://uploads-ssl.webflow.com/60fd4503684b466578c0d307/61138924626a6981ee09caf6_jurassic_tech_paper.pdf	Lieber et al. (2021)	12
GPT-3	https://arxiv.org/pdf/2005.14165.pdf	Brown et al. (2020)	37
InstructGPT	https://arxiv.org/pdf/2203.02155.pdf	Ouyang et al. (2022)	14
Codex	https://arxiv.org/pdf/2107.03374.pdf	Chen et al. (2021)	1
GLM	https://arxiv.org/pdf/2210.02414.pdf	Zeng et al. (2022)	29
Cohere	https://docs.cohere.ai/docs/introduction-to-large-language-models	Cohere	0
ELMo	https://arxiv.org/pdf/1802.05365.pdf	Peters et al. (2018)	6
BERT	https://arxiv.org/pdf/1810.04805.pdf	Devlin et al. (2019)	10
RoBERTa	https://arxiv.org/pdf/1907.11692.pdf	Liu et al. (2019)	11
BART	https://arxiv.org/pdf/1910.13461.pdf	Lewis et al. (2020a)	6
DeBERTa	https://arxiv.org/pdf/2006.03654.pdf	He et al. (2021)	19
FLAN	https://arxiv.org/pdf/2109.01652.pdf	Wei et al. (2022a)	63
GLaM	https://arxiv.org/pdf/2112.06905.pdf	Du et al. (2021)	25
LaMDA	https://arxiv.org/pdf/2201.08239.pdf	Thoppilan et al. (2022)	0
PaLM	https://arxiv.org/pdf/2204.02311.pdf	Chowdhery et al. (2022)	262
Gopher	https://arxiv.org/pdf/2112.11446.pdf	Rae et al. (2021)	20
RETRO	https://arxiv.org/pdf/2112.04426.pdf	Borgeaud et al. (2022)	6
Chinchilla	https://arxiv.org/pdf/2203.15556.pdf	Hoffmann et al. (2022)	14
Alexa-LM	https://arxiv.org/pdf/2208.01448.pdf	Soltan et al. (2022)	27
SuperGLUE	https://arxiv.org/pdf/1905.00537.pdf	Wang et al. (2019a)	8
LM Eval Harness	https://github.com/EleutherAI/lm-evaluation-harness	Gao et al. (2021b)	71
BIG-Bench	https://arxiv.org/pdf/2206.04615.pdf	Srivastava et al. (2022)	208

Table 12: **Summary of language model evaluations.** We enumerate 33 prominent evaluations of language models, spanning both language modeling efforts and benchmarking efforts. Across all of these, they amount to a total of 405 datasets being used to evaluate language models. Here we provide a summary, whereas on Github we annotate the full 405 $\times$ 33 binary matrix.

we evaluate, as was raised by Brown et al. (2020), since many of them are trained on massive, poorly understood, and inadequately documented corpora, where it is certainly possible that models are trained on test instances. In particular, we lack visibility on the full training procedure for the majority of the models we train, and in fact know almost nothing about the training data or training procedure for some models.

In light of this, we attempt to document known evidence of contamination when possible (see Table 13). In particular, beyond models being exposed to the specific evaluation instances, we also consider models contaminated for our purposes if they are exposed to the evaluation distribution, because this compromises the legitimacy of claims about the *few-shot* capabilities of models, even though this training may be very intentional.[111] Given the evidence we observed, we differentiate **strong** contamination, where the model is extensively exposed to model from the evaluation distribution, from **weak** contamination, where the model only sees a marginal fraction of data from the evaluation distribution. We also note that the strength of this evidence can differ: for example, some of the existing evidence is obtained through heuristic n-gram overlap whereas other is known from different knowledge regarding how the training data was constructed. We encourage model providers of both existing and future models to provide more information on train-test contamination to allow the community to be informed about the validity and legitimacy of specific evaluation results, both for HELM and for other evaluations. For HELM, the evidence of contamination will be tracked on Github.

H Priority system

In total, to provide the desired coverage and holistic understanding, our benchmark requires significant resources to be run in full. Notably, numerous queries must be generated and executed on each language model, each requiring the model to process and/or generate a sequence of tokens. All of this computation can prove to be taxing, especially for large models with specific compute requirements and expensive aggregate

[111]We note this point on intentionality to clarify the term "contamination" may suggest that it was unintentional/unexpected, but we take a consequentialist perspective and are focused on whether the validity of our evaluation will be compromised independent of an intentions.